Question **2**

Complete

Mark 1.00 out of 1.00

Flag question

The following C function takes a singly-linked list as input argument. It modified the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.

```
typedef struct node
{
int value;
struct node *next;
} Node;
Node *mode_to_front (Node *head){
    Node *p, *q;
    if((head = = NULL) || (head -> next = = NULL))
        return head;
    q = NULL;
    p = head;
    while (p ->next ! = NULL)
        q=p;
    p = p->next;
}
```

Question **5**

Complete

Mark 4.00 out of 4.00

Flag question

Write C functions to perform following operations on singly linked list.

1. init() // Initializes a list of characters [1 mark]
2. create() // creates a node of character and returns the newly created node [1 mark]
3. add_beg() // adds the node of character 'A' to the beginning of the list. [1 mark]

Note: Do not write main (). You are supposed to write structure for linked list and the above functions only.

Question **3**

Complete

Mark 1.00 out of 1.00

Flag question

```
typedef struct student {
        int roll_no;
        char name[20];
}student;
```

Write a C statement to declare a pointer 'sptr' of type student.

Answer: student *sptr;

Question **4**

Complete

Mark 4.00 out of 4.00

Flag question

Write C functions to perform following operations on singly linked list.

1. init() // Initializes a list of characters [1 mark]
2. append() // adds a node to the end of the list. [1 mark]
3. remove_at_beg() // deletes the first node from the list. [1 mark]

Note: Donot write main(). Use your own structure for list.

```
        while (p ->next ! = NULL)
                q=p;
        p = p->next;
}
_______________
return head;
}
```

Choose the correct alternative to replace the blank line.

- a. head=p; p->next=q; q->next=NULL
- b. q=NULL; p->next=head; head=p;
- c. q->next=NULL; head=p; p->next=head;
- d. q->next=NULL; p->next=head; head=p;

Define ADT Array (Array of integers).

Write following functions with suitable prototypes for ADT Array:

init () // initializes array

append( ) // to insert an element in the end of the array.

traverse() // to display all the elements of the array.

reverse_even() /* (4 Marks)

This function reverses all consecutive even integers in the array.

For example, if the array is: {1, 2, 8, 9,12,16,18,11,14,13}

after the function call the array should changes to:

{1, 8, 2, 9, 18, 16, 12, 11, 14, 13}

*/

You are free to include more functions.

Question 1

Complete

Mark 4.00 out of 4.00

Flag question

Define ADT Array (Array of integers)

Write following functions with suitable prototypes for ADT Array:

init_array() // (1 Mark) initializes the array

append( ) // (1 Mark) to add an element at the end of the array.

sort() // (1 Mark) sorts array in ascending order

traverse() // (1 Mark) to display all the elements of the array starting from last element to first.

Note: Donot write main(). Write your structure for ADT Array